

Software Installation & Environment Setup

Practical Thermodynamic Modelling for Petrology

Ondrej Lexa

Institute of Petrology and Structural Geology (IPSG)
Charles University, Prague, Czechia

GICA2026 Workshop

Welcome

Practical Thermodynamic Modelling for Petrology

A 1-day, hands-on workshop for advanced students, postdocs, and professionals working in metamorphic, igneous, and economic geology.

Core stack for today:

- Python — petropandas, pypsbuilder
- Julia / C — MAGEMin, MAGEMinApp
- Jupyter notebooks throughout

All software is installed live during the session — just bring a laptop with an internet connection.

Today's Schedule

Time	Session	Tools
09:00–10:30	Software installation & setup	petropandas, pypsbuilder, MAGEMinApp
10:30–10:45	<i>Morning coffee break</i>	
10:45–12:30	Data processing	petropandas
12:30–13:30	<i>Lunch break</i>	
13:30–15:30	P–T pseudosections	pypsbuilder / THERMOCALC
15:30–15:45	<i>Afternoon coffee break</i>	
15:45–16:30	Fast minimization	MAGEMin / MAGEMinApp
16:30–17:00	Q&A & wrap-up	

One thread runs through the whole day: raw microprobe data → calibrated bulk composition → P–T path.

This Session

Objective

Get all three software stacks installed and working *before* the hands-on modules begin, so Modules 1–3 can start straight into the science.

- Python environment: petropandas & pypsbuilder (Modules 1–2, one shared install)
- THERMOCALC setup for pypsbuilder (Module 2)
- Julia, MAGEMin & MAGEMinApp (Module 3)

Full copy-pasteable commands are also in `install.md` in the workshop repository — work at your own pace, ask for help anytime.

All workshop materials (slides, notebooks, data):

github.com/ondrolexa/GICA2026_workshop

Python Environment

Install: petropandas & pypsbuilder

Requires Python ≥ 3.12 (workshop targets 3.14). Both packages install together into **one shared project** using uv.

Option A — from scratch

```
uv init --python 3.14 --bare workshop
cd workshop
uv add petropandas[lab] pypsbuilder[pyqt6]
```

Option B — from downloaded workshop materials

unzip, open a terminal in that folder, then:

```
uv sync
```

Then run everything via `uv run` (e.g. `uv run jupyter lab`, `uv run tcinit`).

THERMOCALC

Set Up THERMOCALC

`pypsbuilder` is a front-end — it does not bundle THERMOCALC itself.

Automated (recommended): run in the workshop directory:

```
uv run tcinit
```

Enter `avgpelite` as project directory, `demo` as project name, `4.9, 12.1, 595, 725` as `p–T` range, and choose the `metapelite` set — downloads THERMOCALC 3.51 for your platform, pre-tagged and ready for `ptbuilder`/`txbuilder`/`pxbuilder`.

Manual fallback: prepare THERMOCALC/`drawpd` executables, a preferences file, a dataset/`a-x` file, and insert the `PSBGUESS`/`PSBCALC`/`PSBBULK` scriptfile tags by hand — see `install.md` for the full procedure and example scriptfiles.

MAGEMin / MAGEMinApp

Install: Julia & MAGEMinApp

Requires a recent Julia install.

Launch Julia with all available threads:

```
julia -t auto # or e.g. `julia -t 6`
```

In the Julia REPL:

```
using Pkg
Pkg.add("MAGEMinApp")

using MAGEMinApp
App()
```

Julia prints `Listening on: 127.0.0.1:8050` — open that address in a web browser. `MAGEMin_C` installs automatically as a dependency; no Julia scripting required for this workshop.

Recap

Checklist

- ☐ petropandas installed and importable
- ☐ pypsbuilder installed, THERMOCALC set up via tcinit
- ☐ Julia running, MAGEMinApp installed, App() opens in the browser

Stuck on something? Full copy-pasteable commands and troubleshooting notes live in `install.md` — flag down a helper any time during the day.

Next: Data Processing with petropandas